# Exploring the Effect of Information Sampling on the Relationship Between Prospective Intolerance of Uncertainty and Decision Confidence

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**Abstract:**

This paper investigates the relationship between Prospective Intolerance of Uncertainty (PIU) and self-reported confidence, incorporating the influence of key additional variables. Utilizing descriptive analysis and applying two analytical approaches, Multilevel Structural Equation Modeling (MSEM) and Mediation Analysis ,this study examines whether the relationship between the two variables is mediated by a third variable related to Information Sampling. The dataset used for this study was thoroughly cleaned and free of missing values, ensuring the robustness and reliability of the analysis. The results of both analyses reveal a partial mediation, suggesting that Information Sampling plays a significant but not exclusive role in explaining the relationship between PIU and Confidence.
Note: All the relevant code used for this study is given in the Appendix section of the paper.

**Keywords:** Prospective Intolerance of Uncertainty (PIU) • Confidence • Information Sampling • Decision-making.

# Introduction

Intolerance of Uncertainty (IU) is a psychological characteristic that significantly influences individuals' reactions to uncertain or ambiguous circumstances. This trait is strongly related to anxiety disorders, obsessive-compulsive symptoms, and generalized worry, since IU enhances the view of uncertainty as both threatening and challenging to cope with (Carleton, 2016). IU has been categorized into two primary subfactors: Prospective IU (PIU) and Inhibitory IU (IIU). PIU is a cognitive tendency to decrease the uncertainty about future events. It often manifests itself in information-seeking behavior, whereas IIU is characterized by behavioral restraint and the preference to avoid uncertain situations (Freeston et al., 1994). The graph 1, shows a clear explanation of the characteristic of each type of Intolerance.

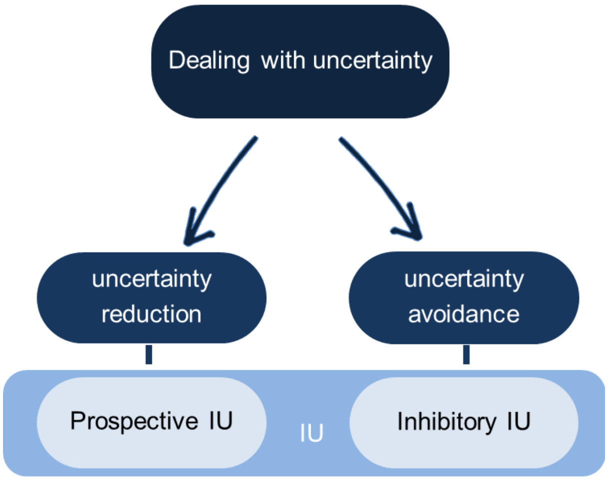



**Figure 1:** Prospective and Inhibitory Intolerance explanation

Decision-making under uncertainty is another important domain that has been influenced by IU. For example, compulsive personality traits are characterized by abnormalities in perceptual decision making as well as metacognitive functions, reflecting impaired abilities to adequately use information in guiding choices (Hauser et al., 2017). Similarly, information-sampling behaviors are embedded within the broader social and cognitive contexts. A study by Ma et al. (2016) has shown that the acquisition of information in trust-based decisions is socially costly and explored how uncertainty and interpersonal considerations influence information-seeking behavior. Developmental differences in information sampling are also important: for instance, teenagers are willing to sample more information compared to adults, especially when decision-making is associated with higher effort costs (Niebaum et al., 2019). These findings point to the complexity of how individuals manage uncertainty in diverse contexts.

In a previous analysis, the relation between PIU and confidence was examined by using a multilevel model, which showed that individuals with higher PIU sampled more information, which in turn boosted their decision confidence. This analysis also included IIU as a control variable; however, because of the high correlation between PIU and IIU, it was very difficult to unravel their distinctive effects. This high multicollinearity—two predictors being highly correlated—is problematic because it inflates standard errors, reduces the statistical power to detect individual predictor effects, and makes coefficient estimates unstable and sensitive to minor changes in the data (Kutner et al., 2004)

The motivation of this work lies in how it contributes to our knowledge of the cognitive and behavioral mechanisms associated with IU. By focusing specifically on PIU, it builds on previous research that links IU to decision-making difficulties in both clinical and non-clinical populations (Dugas et al., 2005). Additionally, examining the ways information sampling mediates this relationship sheds light on the dynamic processes that go into developing confidence. Along with improving our knowledge of IU, these discoveries also have practical implications, such as helping the development of specialized interventions to support individuals struggling with uncertainty.

The present study addresses this limitation by investigating PIU exclusively, thereby allowing for a clearer examination of its unique association with decision-making behaviors. Specifically, it investigates whether the relationship between PIU and self-reported confidence is mediated by the amount of information individuals sampled before making a decision. Figure 2 shows the graph which represent the relationship of these three variables.

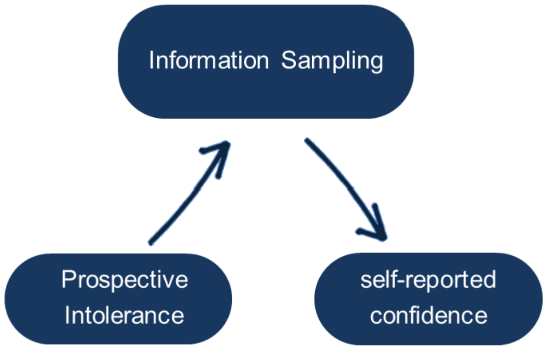



**Figure 2:** Relationship between PIU, COnfidence Ratings and Information Sampling

# Methods

## Sampling Method and Participants

To collect data for this study, 214 participants played a "game" 60 times each. In each trial, participants see a 5x5 grid of gray squares, which can be flipped by clicking to reveal one of the two colors, indicated at the bottom of the screen. The goal is to decide which color is the majority underlying the tiles. Participants can click on any number of tiles during the trial, and the flipped squares remained revealed throughout the trial. At any point, participants can select the majority color by clicking one of the two colors indicated at the bottom of the screen. Once they were confident and made a majority color decision, they had to rate their confidence in the decision between 0 and 100, where 0 means no confidence, and 100 means complete certainty. Each trial has a different distribution of the two colors, with the majority color ranging from 13 to 20 tiles. After completing each trial, participants immediately move on to the next one without knowing whether their answer to the previous trial was correct, creating a continuous sequence of 60 trials per participant. The key variables recorded during each trial included the number of tiles clicked, the participant's confidence rating, whether their final decision was correct, and many others. Figure 3 provides a visual illustration of the experiment to clarify the design of the game and make it easier to understand.

Before completing the information sampling task, participants completed the Intolerance of Uncertainty Scale (IUS) online via Qualtrics. The IUS measures individual differences in prospective intolerance of uncertainty, with scores derived from the sum of relevant items on the scale. For analysis, these scores were standardized by applying z-scoring, a statistical technique that transforms raw scores into values that indicate how many standard deviations a score is from the mean of the dataset. This ensures comparability between participants by placing all scores on a common scale with a mean of 0 and a standard deviation of 1 (Field, 2013).


**Figure 3:** Visual representation of the sequential steps of the game over time.

For this analysis, the focus is on the following variables: **Prospective Intolerance of Uncertainty (PIU)**, **Correct choice**, **Sampling Information**, **Confidence Scaled**, **Trial**, **Response time**, **Age**, **Gender** . These variables were selected because the aim of this project is to investigate the relationships between these specific variables, and including additional variables could introduce too much noise, making it difficult to understand the true relationships among the selected variables.

## Measurements

Moving on, a thorough synopsis of all the factors used in the research is provided. The intention is to clarify the definitions of these variables and describe the methods used for working with them. We start with the variables that are consistent in each trial and one of this is **Age**, which denotes the age of the participants expressed in years. As we examine from the descriptive analysis we did, the age range of the participants is 19-40. Additionally the **Gender** indicate the biological gender of each participant with 1 meaning men and 2 meaning women. The third variable is **PIU** (Prospective Intolerance of Uncertainty), which reflects the tendency to feel anxious about future uncertain events. This variable was scaled by the researcher as described above, with values ranging from -2.7 to 2.7. Higher values indicate that the participant has a stronger tendency to seek out more information, while lower values suggest a preference to avoid learning information.

Continuing with the description of the data used in this study, the variable **Confidence Ratings** was assessed in each trial for every participant. This dependent variable reflects the level of confidence participants felt when making their decision on a given trial, reported on a scale from 0 to 100. From the descriptive analysis, we observed that the mean confidence level was consistent across all age groups and genders, with an average of approximately 85. Additionally, a notable number of confidence ratings clustered at 50 and 100.

Moreover, several other variables related to the "game" and the participants' engagement with it were included, such as **Correct Choice**, **Sampling Information**, **Trial**, and **Response Time**. The variable **Correct Choice** indicates whether the participant's answer in each trial is correct (1 for correct, 0 for incorrect). **Sampling Information** reflects the number of tiles each participant clicked in a given trial. From the descriptive analysis, we observed that this number fluctuated, with participants not follow a specific pattern. The variable **Trial** identifies the sequence of the trials each participant completed, while **Response Time** measures the number of seconds taken by a participant to report their confidence in each trial. These variables are considered critical for addressing our research question, as they provide insights into the factors that influence participant behavior and decision-making processes. Table 1 provides an example of the variables used.

| Profile | 1 | 1 | 1 |
|---|---|---|---|
| Sampling Information | 11 | 10 | 13 |
| PIU scaled | -0.675 | -0.675 | -0.675 |
| Confidence scaled | 96 | 99 | 95 |
| Correct choice | 0 | 1 | 1 |
| Trial | 56 | 19 | 20 |
| Response time | 1044 | 761 | 979 |
| Age | 35 | 35 | 35 |
| Gender | 2 | 2 | 2 |

**Table 1:** Example Data

## Statistical analysis

Based on the descriptive analysis and the nature of the dataset, we decided to focus on two different analyses. The first one is the Multinomial Mediation Analysis, while the second one is the Multilevel Structural Equation Modeling. These methods enable us to address distinct research questions and take advantage of the hierarchical structure and categorical nature of our data.

### Multilevel Structural Equation Modeling

Multilevel Structural Equation Modeling is well-suited for datasets with hierarchical structures as it takes into consideration interdependencies across various levels of analysis (Preacher et al., 2010) ,distinguishing between within-subject (Level 1) and between-subject (Level 2) factors.

A major advantage of MSEM over traditional multilevel mediation is the ability to incorporate latent variables. Latent variable modeling allows one to include unobserved constructs, which may be indirectly observed through measured indicators (Kline, 2015). However, this method also comes with a limitation. Multicollinearity among the predictors is still a concern, as it can complicate the interpretation of the individual effects.

This method is really relevance to our research question, which investigates how the relationship between Prospective Intolerance of Uncertainty (PIU) and self-reported decision confidence is moderated by the amount of information that people seek before making a decision. A multilevel structural approach will let us examine how changes in the mediators affect. Our study topic analyze and compare two different models. The first one, focuses on how the amount of information sought before making a decision influences the link between Prospective Intolerance of Uncertainty (PIU) and self-reported decision confidence and the secodn one how Prospective Intolerance of Uncertainty (PIU) and self-reported decision confidence connected directly .

### Multilevel mediation analysis

Traditional mediation analysis (Baron and Kenny, 1986) tries to explain how an independent variable influences a dependent variable through the inclusion of one or more mediating variables. The concept of multilevel mediation analysis extends this paradigm to cases in which the dependent variable is categorical and includes three or more levels. This is of particular relevance to our research question, which investigates how the relationship between Prospective Intolerance of Uncertainty (PIU) and self-reported decision confidence is moderated by the amount of information that people seek before making a decision. A multilevel approach will let us examine how changes in the mediators affect the likelihood of each categorical outcome, therefore giving a nuanced understanding of the decision-making process.

Multilevel mediation analysis is a powerful tool but has its inherent problems. The most striking issue is probably multicollinearity between predictors—meaning highly correlated independent variables, which complicates estimation. This multicollinearity can result in inflated standard errors, unstable coefficient estimates, and problems with interpreting the results for single predictors (Dormann et al., 2013). Additionally, the correlation between observations obtained from the same participant precludes the use of standard analysis methods. The clustered nature of data within participants must be recognized as a key reason for adopting a multilevel structure.

Furthermore, while multilevel models assume independence across outcome categories—the so-called independence of irrelevant alternatives—this assumption differs from the commonly understood independence of distinct observations. In this context, participants' lack of knowledge about whether their answers in each trial were correct is crucial, as it preserves the independence of their decision-making across trials. However, in practice, violations of the independence of irrelevant alternatives assumption can still occur, potentially leading to biased results (Long  Freese, 2014). Multilevel mediation analyses thus empower us to tease apart complex relationships in great detail when used judiciously.

**Software**

The R statistical programming language, along with relevant packages so we can do multilevel structural equation modeling and multilevel mediation analysis. The packages were used are lavaan (Yves Rosseel et al., 2012), lme4 (Douglas Bates et al., 2015) and psych (William Revelle el at., 2024. R was chosen since it offers a stable and adaptable frame work for statistical analysis that guarantees the results' reproducibility and transparency.

# Results

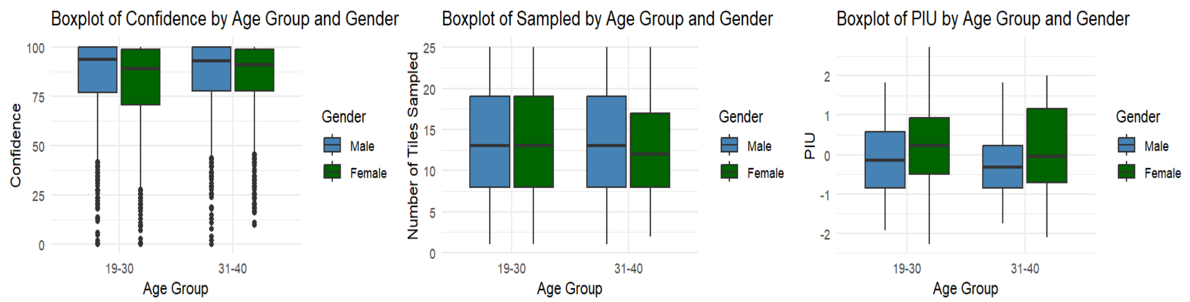
### Descriptive Analysis

We first explored the descriptive statistics of the variables PIU, Confidence, and sampled data, along with age and gender, as it is useful to conduct an initial exploration that may reveal interesting insights. The age of participants ranges from 19 to 40, so we decided to divide them into two groups (19-30 and 31-40) as participants in these age ranges tend to show similar patterns. From Table 2, we can see that men and women are equally distributed across all age groups. However, men are generally more confident than women, regardless of their age. Women have a higher PIU value, which could mean that they want to learn more information compared to men. This is interesting since the number of tiles opened by each gender seems to go against their PIU values: men open more tiles despite having a lower PIU.

| Age | Gender | Count | Confidence | SD | Sampled | SD | PIU | SD |
|---|---|---|---|---|---|---|---|---|
| 19-30 | Men | 3299 | 85.70 | 17.93 | 13.82 | 6.69 | 0.15 | 0.89 |
| 19-30 | Women | 3660 | 82.94 | 18.69 | 13.37 | 6.79 | 0.19 | 1.03 |
| 31-40 | Men | 2700 | 86.75 | 15.96 | 13.59 | 6.63 | -0.19 | 0.90 |
| 31-40 | Women | 3120 | 86.04 | 15.98 | 12.92 | 6.31 | 0.07 | 1.06 |

**Table 2:** Descriptive analysis of confidence ratings (Confidence), the number of tiles sampled (Sampled), and Prospective Intolerance of Uncertainty (PIU) by age group and gender. The table includes the mean values for each variable, with the corresponding standard deviation (SD) listed for confidence, sampled, and PIU.

To help understand the key variables better, the boxplots are shown in Figure 4.




(a) (b) (c)

**Figure 4:** Boxplots of confidence, number of tiles sampled, and prospective intolerance of uncertainty (PIU) across participants. Panel (a) shows the distribution of confidence ratings (Confidence), panel (b) represents the number of tiles sampled during each trial (Sampled), and panel (c) displays the distribution of prospective intolerance of uncertainty (PIU). The plots illustrate the variations across different participant characteristics.

The analysis also revealed that both mean confidence and mean number of tiles sampled across trials do not follow a linear pattern. As it is clear from Figure 5, the lines showed fluctuations, going up and down across the trials. To address this non-linearity, both a mediator and an outcome model were fitted using the original variables, as well as incorporating a cubic transformation of the trial variable. This approach was aimed at testing whether accounting for potential non-linear relationships could improve the model fit.

To compare the two models, an ANOVA test was conducted. Analysis of Variance (ANOVA) is a statistical method used to compare models by assessing differences in their goodness of fit while penalizing for model complexity (Akaike, 1974). It evaluates metrics such as the Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), log-likelihood, and deviance to determine if one model is statistically superior to another. In our case, the cubic models had slightly better AIC, BIC, and deviance values than the linear models, suggesting a marginally improved fit. However, the differences were very small, and the Chi-Square test indicated that the improvement was not statistically significant. Given this outcome, we opted to retain the linear models to avoid overfitting, as the additional complexity of the cubic models was not justified by a meaningful improvement in model performance. This decision aligns with the principle of parsimony, which emphasizes simplicity in model selection unless complexity offers clear advantages.

The results showed that the differences in AIC, BIC, and deviance were minimal, and the Chi-Square test indicated that the improvement in fit was not statistically significant. Given the small difference between the models and to avoid overfitting, the decision was made to retain the linear model. The lack of a substantial or statistically significant improvement in fit did not justify the added complexity of the cubic model.

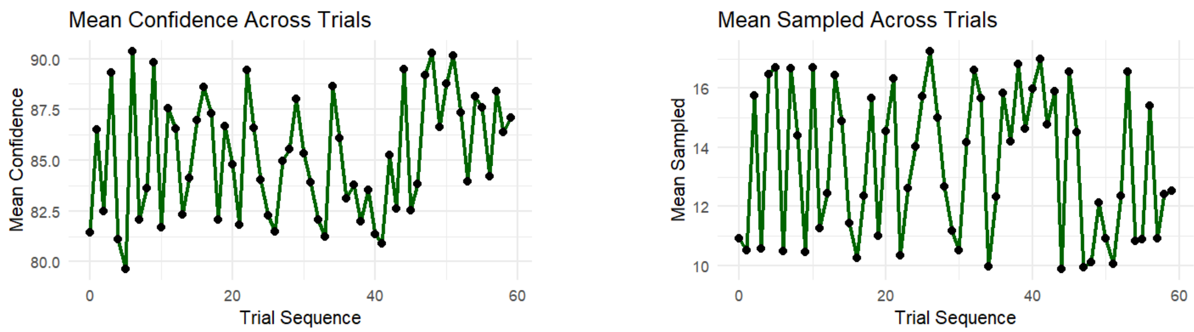



**Figure 5:** Mean confidence and mean sampled across trials

## Multilevel Structural Equation Modeling

In this analysis, we fitted 2 models. In the Model 1, we focused on examining the mediation effect we were interested in while in Model 2, we focused on understand the direct effect of PIU to Confidence. In Level 1 of Model 1, the outcome variable, Confidence, was modeled as being influenced by several within-subject predictors: the number of tiles sampled, Response Time, trial sequence, and whether the participant made a correct choice. At Level 2 of Model 1, we explored how between-subject variables, such as PIU, Age, and Gender, affect the number of tiles sampled. This allowed us to investigate whether the number of tiles sampled mediates the relationship between Confidence and PIU. In Model 2, we conducted a forward analysis to examine the relationship between Confidence and the other predictors. Specifically, we tested the direct effect of Confidence on PIU to understand better how these variables are related. To ensure the reliability of our mediation estimates, we employed bootstrapping for model fitting, which helped to estimate the indirect and direct effects of PIU to Confidence. The Figure 6 indicates an illustration of how Model 1 and Model 2, built.

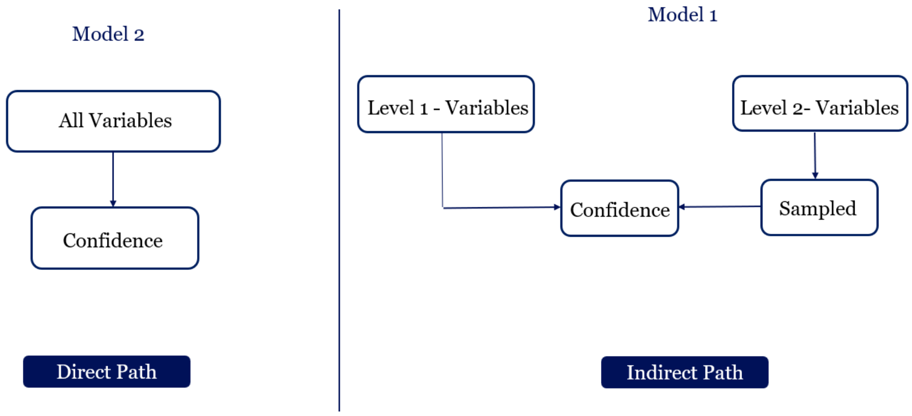



**Figure 6:** Visualization of the two model in Multilevel Structural Analysis

Regarding the Model 1, the Tables 3, 4 show the influence of each variable in Confidence (Level 1) and influence of the other variables in Information Sampling (Level 2) respectively. Firstly we can conclude that all predictors are important and influence Confidence since their P-value is less than 0.05. Based on the Table 3, the most important variable that influence Confidence is Correct Choice . When the Correct Choice increase 1 sd, the confidence increase 17.8 sd. Based on the Table 4 we can observe a negative association between Gender, PIU on Information Sampling.

Regarding the Model 2, the Table 5 shows the influence of each variable on Confidence. Firstly we can observe that all predictors are important and influence Confidence since their P-value is less than 0.05. The exact p-values are really equally to 0, so we decided to present them in this format. The most important variable is Correct Choice . When the correct choice increase 1 sd, the confidence increase 17.8 units. Additionally there is negative association between Gender and Confidence. The Number of the Trial and the Response time, do not look as important as the other variables.

| | **Estimate** | **P-value** |
|---|---|---|
| Sampled | 0.92 | <0.05 |
| Response time | 0.003 | 5.7e-08 |
| Number of Trial | 0.04 | 5.7e-05 |
| Correct Choice | 17.8 | <0.05 |

**Table 3:** Model 1 - Level 1 results

| | **Estimate** | **P-value** |
|---|---|---|
| Age | 0.06 | 3.58e-08 |
| Gender | -0.07 | 2.9e-12 |
| PIU | -0.75 | <0.05 |

**Table 4:** Model 1 - Level 2 results

| | **Estimate** | **P-value** |
|---|---|---|
| Sampled | 0.91 | <0.05 |
| Response Time | 0.003 | <0.05 |
| Number of Trial | 0.04 | <0.05 |
| Correct Choice | 17.75 | <0.05 |
| PIU | 1.09 | <0.05 |
| Age | 0.21 | <0.05 |
| Gender | -1.59 | <0.05 |

**Table 5:** Model 2 results

The Table 6 indicates the evaluation metrics of these two models with which we can draw some conclusions. Overall, while both models show similar fit statistics, the very small difference in BIC values might suggest a preference for the second model. Specifically, BIC balances the trade-off between a model's quality of fit and complexity , so it is important to be low.

Finally ,using the previous results, we focus to determine the impact of PIU on Confidence, either through Information Sampling or directly. Table 7 shows that the amount of information sampled plays a role in the relationship between PIU and Confidence, as the p-value is less than 0.05 and the estimate is 0.58. At the same time, PIU has a strong direct effect on Confidence, with an estimate of 1.09.

| | Model 1 | Model 2 |
|---|---|---|
| CFI | 0.99 | 0.99 |
| TLI | 0.97 | 0.96 |
| AIC | 190 976.5 | 190 857.7 |
| BIC | 191 058.6 | 190 962.1 |

**Table 6:** Evaluation Metrics

| | Estimate | P-value |
|---|---|---|
| Direct | 1.09 | <0.05 |
| Indirect | 0.58 | <0.05 |

**Table 7:** Direct and Indirect path

**Multilevel Mediation Analysis** In this analysis, the modeling process involves building two models, each testing different hypotheses between variables.

1. **Mediator Model** This model investigates the relationship between Prospective Intolerance of Uncertainty (PIU) and the mediator variable, which is the amount of information sampled before making a decision. This will allow us to examine whether PIU has a significant effect on information sampling or not.

2. **Outcome Model** This model examines the relationship between the mediator (information sampling) and the outcome variable, which is confidence in decisions while taking into consideration the direct effect of PIU on confidence. This will allow us to better understand if and how much PIU impacts confidence through information sampling.

Both models include Response Time, Count Trial, Correct Choice, Age, and Gender as control variables to account for potential confounding. A visualization of the two models described above is presented in Figure 7.

The findings from both models will provide a comprehensive understanding of the relationships between PIU, information sampling, and confidence. The mediator model will shed light on how PIU might influence the amount of information sampled, while the outcome model will explore how information sampling, influenced by PIU, ultimately affects confidence. This approach helps to answer the research question by clarifying both the direct and indirect pathways through which PIU impacts decision-making confidence.

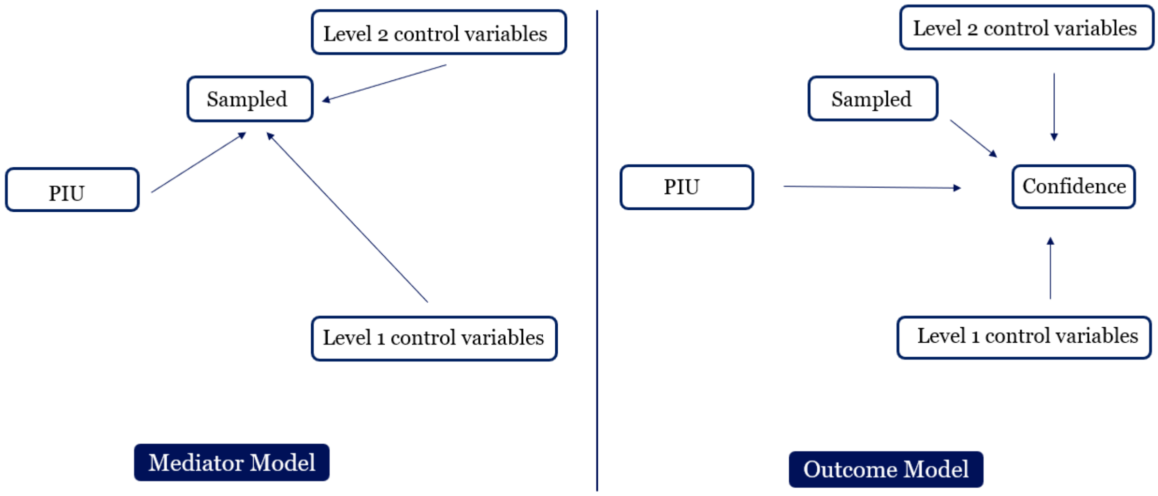



**Figure 7:** Visualization of the two models

The results of the mediator models show several significant relationships. PIU, has a positive, significant effect on information sampling (T = 1.83), implying that people with higher PIU tend to gather more information. Response time seems to have a strong, positive relationship with information sampling (T = 27.69), which is expected as it means that the longer someone takes to respond the more information they need to sample. The number of trials and correct choice have a negative relationship with sampling ( T = -3.34 and T = -8.52 respectively), suggesting that participants tend to sample less information as they progress through the trials and that they tend to answer incorrectly when sampling less information. Both Age (T = -0.99) and Gender (T = -1.127) show negative but no significant association with the information sampling. The estimates of the variables and the T-value are presented in the table 8.

| **Predictor** | **Estimate** | **T-value** |
|---|---|---|
| Response Time | 0.162 | 27.687 |
| Number of Trials | -0.019 | -3.339 |
| Correct Choice | -0.210 | -8.521 |
| PIU | 0.094 | 1.825 |
| Age | -0.050 | -0.994 |
| Gender | -0.116 | -1.127 |

**Table 8:** Predictors of Information Sampling in the Mediator Model

To evaluate the two models, we computed AIC, BIC. The mediator model has a slightly better balance of fit and complexity with lower AIC (21,597.181) and BIC (21,679.19) compared to the outcome model.

| Metric | Mediator Model | Outcome Model |
|---|---|---|
| AIC | 21,597.18 | 24,656.69 |
| BIC | 21,679.19 | 24,746.15 |

**Table 9:** Model Fit Indices and Mediation Results

We then proceed with the Mediation Analysis to address the research question. Firstly, in order to ensure that our mediation analysis results are robust and generalizable, we conducted a bootstrap analysis with 1000 iterations, which will provide more reliable estimates and confidence intervals as it keeps recalculating these effects by resampling the data (Efron, 1979).

The results show that the effect of PIU on information sampling is positive as the mean is positive. Furthermore, the direct effect, which represents the effect of PIU on confidence that is not mediated by information sampling, and the total effect, which is the overall impact of PIU's direct and indirect (via information sampling) effects on confidence, are also positive. None of the confidence intervals include zero, which strongly indicates that both the indirect and direct effects are statistically significant.

The findings show partial mediation, as the direct effect (0.0822) is larger than the indirect effect (0.0161), which indicates that most of the effect of PIU on confidence operates directly, rather than through information sampling.

| Effect | Mean | 95% Confidence Interval |
|---|---|---|
| Indirect Effect | 0.0161 | [0.0131, 0.0190] |
| Direct Effect | 0.0822 | [0.0721, 0.0925] |
| Total Effect | 0.0983 | [0.0882, 0.1086] |

**Table 10:** Bootstrap Results for Mediation Analysis (1,000 Iterations)

# Discussion

In this study ,the goal is to investigate whether the relationship between PIU and self-reported confidence is mediated by the amount of information individuals sampled before making a decision. We address the research question using two analyses: Multilevel Structural Equation Modeling and Mediation Analysis.

Multilevel Structural Modeling, fits 2 models. Model 1, have 2 levels and it focus on examining the mediation effect (Information Sampling) between Intolerance of Uncertainty (PIU) to Confidence in decisions ,while Model 2, focus to explore the direct effect of Prospective Intolerance of Uncertainty (PIU) to Confidence in decisions.

Mediation Analysis, builds also 2 models. The mediator and the outcome. The Mediation model Mediator Model investigates the relationship between Prospective Intolerance of Uncertainty (PIU) and the mediator variable, while the Outcome examines the relationship between the mediator (information sampling) and the outcome variable, which is confidence in decisions while taking into consideration the direct effect of PIU on confidence.

By combining the results of both models we were able to draw some conclusions and answer the research question. The Multilevel Structural Equation Model and Mediation Analaysis Model, suggests that most of the effect of PIU on Confidence comes directly, rather than through Information Sampling, so there is partial mediation.

# Future Work

While this research provides valuable insights and important information, there are some additional changes that can be done in order to draw more general conclusions. As noted earlier, this analysis focused only on Prospective Intolerance (PIU) rather than Inhibitory Intolerance (IU) due to the high correlation between the two and the limitations of the two models which can not handle correlated predictors. A possible solution to address this limitation is to combine the two types of Intolerance using dimensionality reduction techniques such as Principal Component Analysis (PCA), in order to draw more general conclusions.

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# Appendix

R Code:

```
library(dplyr)
library(tidyr)
library(ggplot2)
library(psych)
library(corrplot)
our_data = read.csv('data_all_clean.csv')
```

```r
# General Statistics for the response variable
(scaled_IUS_prospective) and the predictors

our_data %>%
  summarise(
    mean_IUS_prospective = mean(scaled_IUS_prospective, na.rm =
       TRUE),
    sd_IUS_prospective = sd(scaled_IUS_prospective, na.rm = TRUE),
    mean_sampled = mean(sampled, na.rm = TRUE),
    sd_sampled = sd(sampled, na.rm = TRUE),
    mean_confidence = mean(x_AG1_scaled, na.rm = TRUE),
    sd_confidence = sd(x_AG1_scaled, na.rm = TRUE)
  )

# Create age groups
data = our_data %>%
  mutate(AgeGroup = cut(AgeInYears,
                        breaks = c(0, 18, 30, 40, 50, 60, 70, Inf)
                           ,
                        labels = c("0-18", "19-30", "31-40",
                        "41-50", "51-60", "61-70", "70+"),
                        right = FALSE))

# Calculate mean of x_AG1_scaled by age group and gender
age_gender_stats = data %>%
  group_by(AgeGroup, Gender) %>%
  summarise(
    Count = n(),
    MeanConfidence = mean(x_AG1_scaled, na.rm = TRUE),
    MeanSampled=mean(sampled,na.rm=TRUE),
    MeanPIU=mean(scaled_IUS_prospective,na.rm=TRUE),
    .groups = 'drop'
  )

# Print the summary table
print(age_gender_stats)

# Create age groups
```

```r
data = our_data %>%
  mutate(AgeGroup = cut(AgeInYears,
                        breaks = c(0, 18, 30, 40, 50, 60, 70, Inf)
                           ,
                        labels = c("0-18", "19-30", "31-40",
                        "41-50", "51-60", "61-70", "70+"),
                        right = FALSE))

# Calculate mean and standard deviation of x_AG1_scaled by age
group and gender

age_gender_stats = data %>%
  group_by(AgeGroup, Gender) %>%
  summarise(
    Count = n(),
    MeanConfidence = mean(x_AG1_scaled, na.rm = TRUE),
    SDConfidence = sd(x_AG1_scaled, na.rm = TRUE),
    MeanSampled = mean(sampled, na.rm = TRUE),
    SDSampled = sd(sampled, na.rm = TRUE),
    MeanPIU = mean(scaled_IUS_prospective, na.rm = TRUE),
    SDPIU = sd(scaled_IUS_prospective, na.rm = TRUE),
    .groups = 'drop'
  )

# Print the summary table
print(age_gender_stats)

data$Gender <- factor(data$Gender, levels = c(1, 2),
                      labels = c("Male", "Female"))

custom_colors <- c("Male" = "#4682B4", "Female" = "#006400")

# boxplot for confidence (x_AG1_scaled) by age group and gender
ggplot(data, aes(x = AgeGroup, y = x_AG1_scaled, fill = Gender)) +
  geom_boxplot() +
  scale_fill_manual(values = custom_colors) +
  labs(title = "Boxplot of Confidence by Age Group and Gender",
       x = "Age Group",
       y = "Confidence") +
```

```
  theme_minimal()

# boxplot for sampled by age group and gender
ggplot(data, aes(x = AgeGroup, y = sampled, fill = Gender)) +
  geom_boxplot() +
  scale_fill_manual(values = custom_colors) +
  labs(title = "Boxplot of Sampled by Age Group and Gender",
       x = "Age Group",
       y = "Number of Tiles Sampled") +
  theme_minimal()

# boxplot for PIU (scaled_IUS_prospective) by age group and gender
ggplot(data, aes(x = AgeGroup, y = scaled_IUS_prospective,
                 fill = Gender)) +
  geom_boxplot() +
  scale_fill_manual(values = custom_colors) +
  labs(title = "Boxplot of PIU by Age Group and Gender",
       x = "Age Group",
       y = "PIU") +
  theme_minimal()

# Descriptive statistics grouped by subject (PROLIFIC_PIC)
data = our_data %>%
  group_by(PROLIFIC_PID) %>%
  summarise(
    mean_IUS_prospective = mean(scaled_IUS_prospective, na.rm =
       TRUE),
    mean_sampled = mean(sampled, na.rm = TRUE),
    mean_confidence = mean(x_AG1_scaled, na.rm = TRUE)
  )%>%
  arrange(desc(mean_confidence))

# Print the summary table
print(data)

data_PIU = our_data %>%
  group_by(PROLIFIC_PID) %>%
    summarise(
    mean_IUS_prospective = mean(scaled_IUS_prospective, na.rm =
```

```
        TRUE),
     mean_sampled = mean(sampled, na.rm = TRUE),
     mean_confidence = mean(x_AG1_scaled, na.rm = TRUE))%>%
     arrange(desc(mean_confidence))


data_PIU[which(data_PIU$mean_IUS_prospective >= 1.5),]

# 1) Box plot for confidence ratings (x_AG1_scaled) grouped by
whether the final choice was correct (correct_choice)

ggplot(our_data, aes(x = factor(correct_choice), y = x_AG1_scaled)
   ) +
  geom_boxplot(fill = "lightgreen", color = "black") +
  labs(title = "Confidence Ratings by Correct Choice",
       x = "Correct Choice (0 = Incorrect, 1 = Correct)",
       y = "Confidence Ratings (Scaled)") +
  theme_minimal()

# 2) Scatter plot of confidence ratings (x_AG1_scaled) by number
   of
tiles sampled

ggplot(our_data, aes(x = sampled, y = x_AG1_scaled)) +
  geom_point(alpha = 0.5, color = "red") +
  labs(title = "Scatter Plot of Confidence Ratings vs. Number
  of Tiles Sampled",
       x = "Number of Tiles Sampled", y = "Confidence Ratings (
          Scaled)") +
  theme_minimal()

# 3) Plot mean sampled across trials:

# mean confidence for each trial (count_trial_seq)
mean_sampled <- our_data %>%
  group_by(count_trial_seq) %>%
  summarise(mean_samp = mean(sampled, na.rm = TRUE))

ggplot(mean_sampled, aes(x = count_trial_seq, y = mean_samp)) +
```

```
  geom_line(color = "darkgreen", size = 1) +
  geom_point(color = "black", size = 2) +
  labs(title = "Mean Sampled Across Trials",
       x = "Trial Sequence",
       y = "Mean Sampled") +
  theme_minimal()

# 4) Plot mean confidence across trials:


#  mean confidence for each trial (count_trial_seq)
mean_confidence <- our_data %>%
  group_by(count_trial_seq) %>%
  summarise(mean_conf = mean(x_AG1_scaled, na.rm = TRUE))

ggplot(mean_confidence, aes(x = count_trial_seq, y = mean_conf)) +
  geom_line(color = "darkgreen", size = 1) +
  geom_point(color = "black", size = 2) +
  labs(title = "Mean Confidence Across Trials",
       x = "Trial Sequence",
       y = "Mean Confidence") +
  theme_minimal()

# 5) Plot the correlation between numeric variables in the dataset

#Cretae the correaltion matrix
numeric_data = our_data %>% select(scaled_IUS_prospective, sampled
   , correct_choice, x_AG1_scaled)
correlation_matrix = cor(numeric_data, use = "complete.obs")
corrplot(correlation_matrix, method = "circle", type = "upper", tl
   .cex = 0.8)

# 6) Histograms of PIU, sampled and confidence

# Histogram for scaled_IUS_prospective
hist(data$mean_IUS_prospective,
     main = "Histogram of scaled_IUS_prospective",
     xlab = "PIU Score",
     col = "lightblue",
```

```r
     border = "black")

# Histogram for sampled (information sampling)
hist(data$mean_sampled,
     main = "Histogram of Information Sampling (sampled)",
     xlab = "Number of Tiles Turned",
     col = "lightgreen",
     border = "black")


# Histogram for x_AG1_scaled (confidence ratings)
hist(data$mean_confidence,
     main = "Histogram of Confidence Ratings (x_AG1_scaled)",
     xlab = "Confidence (0-100)",
     col = "lightcoral",
     border = "black")

# 7) Visualize the confidence of random people to check if we see
   strange patterns

set.seed(123)  # Set seed for reproducibility (optional)
random_clients = sample(unique(our_data$PROLIFIC_PID),20)

person_sp = function(client){

person = our_data[which(our_data$PROLIFIC_PID == client),
c("scaled_IUS_prospective", "sampled", "x_AG1_scaled", "correct_
   choice", "count_trial_seq")]
person = person %>% arrange(count_trial_seq)
sampled = person$sampled
correct_choice = person$correct_choice
PIU = person$scaled_IUS_prospective[1]



plot = ggplot(data = person, aes(x = count_trial_seq, y = x_AG1_
   scaled)) +
    geom_point() +
    geom_line(color = "blue") +
```

```
    labs(x = "Count_Trial_Seq", y = "x_AG1 Scaled") +
    theme_minimal() +
    ggtitle("Scatter Plot of Count_Trial_Seq vs. x_AG1 Scaled")
  return(list(sampled = sampled, corect_choice = correct_choice,
     PIU = PIU,plot=plot))
}

person_sp("63bd83108d280e9d55d5e33d")
person_sp("64467f4be788c4d0c4d5ff33")
person_sp("63d7f37272879efbbf72dfb0")
person_sp("5a6759ad35f26b0001496694")
person_sp("6400da6aae94a6b881964711")
person_sp("63d3fb12ac52371a49e65860")
person_sp("5e8cb846e94a4406307d6da2")
person_sp("611d2cc49af322d296ac4117")
person_sp("5da41f84a39c100018614e31")
person_sp("6447a4bd54ab5cb19fd4beab")
person_sp("6400d8ee84822ae55ab5c542")
person_sp("5892537355550f0001509c35")

# Mediation Analysis

library(lavaan)
library(lme4)
library(lmerTest)
library(MuMIn)

# standardize the predictor variables so that they are all on the
   same scale

data$scaled_IUS_prospective <- scale(data$scaled_IUS_prospective)
data$response_time_new_mouse_response <- scale(data$response_time_
   new_mouse_response)
data$count_trial_seq <- scale(data$count_trial_seq)
data$correct_choice <- scale(data$correct_choice)

# Model for the mediator (sampled) as a function of the predictor
   and Level 1 and Level 2 control variables (Age, Gender)
```

```
mediator_model_linear <- lmer(sampled ~ scaled_IUS_prospective +
   response_time_new_mouse_response + count_trial_seq + correct_
   choice + AgeInYears + Gender + (1 | PROLIFIC_PID), data = data)
summary(mediator_model_linear)

mediator_model_cubic <- lmer(sampled ~ scaled_IUS_prospective +
   response_time_new_mouse_response + I(count_trial_seq^3) +
   correct_choice + AgeInYears + Gender + (1 | PROLIFIC_PID), data
    = data)
summary(mediator_model_cubic)

anova(mediator_model_cubic, mediator_model_linear)

#  outcome model (x_AG1_scaled) as a function of the mediator (
   sampled), the Level 1 control variables, and Level 2 variables
   (Age, Gender)

outcome_model_linear <- lmer(x_AG1_scaled ~ sampled + scaled_IUS_
   prospective + response_time_new_mouse_response + count_trial_
   seq + correct_choice + AgeInYears + Gender + (1 | PROLIFIC_PID)
   , data = data)
summary(outcome_model_linear)

outcome_model_cubic <- lmer(x_AG1_scaled ~ sampled + scaled_IUS_
   prospective + response_time_new_mouse_response + I(count_trial_
   seq^3) + correct_choice + AgeInYears + Gender + (1 | PROLIFIC_
   PID), data = data)
summary(outcome_model_cubic)

anova(outcome_model_cubic,outcome_model_linear)

# Bootstrap

set.seed(123)

# number of bootstrap samples
n_boot <- 1000

# initialize vectors
```

```
indirect_effects <- numeric(n_boot)
direct_effects <- numeric(n_boot)
total_effects <- numeric(n_boot)

# bootstrapping
for (i in 1:n_boot) {
  # sample with replacement
  boot_data <- data[sample(1:nrow(data), replace = TRUE), ]

  # mediator model
  boot_model_mediator <- lmer(sampled ~ scaled_IUS_prospective +
                                correct_choice +
                              count_trial_seq  +
                              response_time_new_mouse_response +
                                AgeInYears +
                                Gender + (1 | PROLIFIC_PID),
                              data = boot_data)

  # outcome model
  boot_model_outcome <- lmer(x_AG1_scaled ~ sampled + scaled_IUS_
     prospective +
                             correct_choice + count_trial_seq +
                           response_time_new_mouse_response +
                             AgeInYears + Gender +
                             (1 | PROLIFIC_PID), data = boot_data)

  # extract coefficients
  a <- fixef(boot_model_mediator)["scaled_IUS_prospective"]  #
     effect on mediator
  b <- fixef(boot_model_outcome)["sampled"]            # effect of
     mediator on outcome
  c_prime <- fixef(boot_model_outcome)["scaled_IUS_prospective"] #
      direct effect on outcome

  # calculate effects
  indirect_effects[i] <- a * b        # indirect effect
  direct_effects[i] <- c_prime        # direct effect
  total_effects[i] <- indirect_effects[i] + direct_effects[i] #
     total effect
```

```
  if (i %% 10 == 0) {
    cat("Iteration", i, "of", n_boot, "completed\n")
  }
}

# summarize the effects
indirect_summary <- c(mean = mean(indirect_effects), quantile(
   indirect_effects, probs = c(0.025, 0.975)))
direct_summary <- c(mean = mean(direct_effects), quantile(direct_
   effects, probs = c(0.025, 0.975)))
total_summary <- c(mean = mean(total_effects), quantile(total_
   effects, probs = c(0.025, 0.975)))

indirect_summary
direct_summary
total_summary

library(lavaan)
library(psych)
library(ggplot2)
library(semPlot)
library(dplyr)
library(lme4)
library(mediation)


#Read the data
data = read.csv('C:/Users/myrto/OneDrive/Desktop/2nd year/
   Statistical Consulting/statisticalConsulting_docs/data_all_
   clean.csv')
```

Firstly, we are going to modify the dataset and select the
   variables
that we believe are important for the analyses.

```{r}
#Select the variables we decided that are important for the
```

```
    analysis
vars = c("PROLIFIC_PID",
         "sampled",
         "scaled_IUS_prospective",
         "x_AG1_scaled",
         "correct_choice",
         "count_trial_seq",
         "response_time_new_mouse_response",
         "AgeInYears",
         "Gender")

our_data=data[,vars]
#print(our_data)
```

Chosen variables:

-> Profilic_PID = profile of each candidate sampled = number of tiles
-> turned (information sampling) scaled_IUS_prospective = PIU scaled
-> x_AG1_scaled = confidence ratings correct_choice = if their choice was
-> correct in each trial count_trial_seq = number of trial
-> response_time_new_mouse_response = seconds until he/she fill his/her
-> confidence
-> AgeInYears = age
-> gender = gender

1) Multilevel Structural Equation Modeling:

Firstly we built the model with the 2 levels, which shows mediation between PIU and Confidence. This is the indirect path .

```{r}
```

```
# Define the SEM model in lavaan format

model = '
  # Level 1 (Within Subjects)  metaxi trials idiou atomou
  x_AG1_scaled ~ b1 * sampled + b2 * response_time_new_mouse_
     response + b3 * count_trial_seq + b4 * correct_choice

  # Level 2 (Between Subjects) metaxi atoma
  sampled ~ a1 * scaled_IUS_prospective + a2 * AgeInYears + a3 *
     Gender

  # Mediation
  indirect := a1 * b1
'

# Fit the model using lavaan with bootstrapping for indirect
   effects
fit = sem(model, data = our_data, cluster = "PROLIFIC_PID", se = "
   bootstrap", bootstrap = 100)
summary(fit, standardized = TRUE, fit.measures = TRUE)

# Define the SEM model in lavaan format

model_1 = '
  # Level 1 (Within Subjects)  metaxi trials idiou atomou
  x_AG1_scaled ~ b1 * sampled + b2 * response_time_new_mouse_
     response + b3 * count_trial_seq + b4 *correct_choice + b5 *
     scaled_IUS_prospective + b6 * AgeInYears + b7 * Gender


  # Mediation
  indirect := b7
'

# Fit the model using lavaan with bootstrapping for indirect
   effects
fit = sem(model_1, data = our_data, cluster = "PROLIFIC_PID", se =
    "bootstrap", bootstrap = 100)
summary(fit, standardized = TRUE, fit.measures = TRUE, rsquare =
```

```
   TRUE)
estimates = parameterEstimates(fit)
print(estimates)

#comparison between 2 models. For the one the response time is
   linear, for the other is cubic

# Linear Model
model_linear = '
  # Level 1 (Within Subjects) metaxi trials idiou atomou
  x_AG1_scaled ~ b1 * sampled + b2 * response_time_new_mouse_
     response + b3 * count_trial_seq + b4 * correct_choice

  # Level 2 (Between Subjects) metaxi atoma
  sampled ~ a1 * scaled_IUS_prospective + a2 * AgeInYears + a3 *
     Gender

  # Mediation
  indirect := a1 * b1
'

# Cubic Model
model_cubic = '
  # Level 1 (Within Subjects) metaxi trials idiou atomou
  x_AG1_scaled ~ b1 * sampled + b2 * response_time_new_mouse_
     response + b3 * count_trial_seq^3 + b4 * correct_choice

  # Level 2 (Between Subjects) metaxi atoma
  sampled ~ a1 * scaled_IUS_prospective + a2 * AgeInYears + a3 *
     Gender

  # Mediation
  indirect := a1 * b1
'

fit_linear = sem(model_linear, data = our_data, cluster = "
   PROLIFIC_PID", se = "bootstrap", bootstrap = 100)
fit_cubic = sem(model_cubic, data = our_data, cluster = "PROLIFIC_
   PID", se = "bootstrap", bootstrap = 100)
```

```r
anova(fit_linear, fit_cubic)
```